1(b). If $p \in WI_{[E]}(w)$ then by Definition 4.1(2) there exists w'' and v such that $E.p.w'' \simeq E.w.v$. Since $E.w \simeq E.w'$, by Property (6) of Definition 3.2 we also have $E.w.v \simeq E.w'.v$ and transitively $E.p.w'' \simeq E.w'.v$ which by Definition 4.1(2) entails $p \in WI_{[E]}(w')$.

2(a). Trivial, from Property (3) of Definition 3.2 since $E.w$ is a prefix of $E.w.p$ therefore any event not having others happening before it in w maintains this status.

2(b). If $p' \in w$, then $p' \in WI_{[E]}(w.p)$ implies $p' \in I_{[E]}(w)$, which implies $p' \in WI_{[E]}(w)$. If $p' \notin w$, then: if $p' = p$, $p \in WI_{[E]}(w.p)$ implies $E \vdash p \diamond w$ and hence $p \in WI_{[E]}(w)$, otherwise $p' \in WI_{[E]}(w.p)$ implies $E \vdash p' \diamond w.p$ which also implies $E \vdash p' \diamond w$ and therefore $p \in WI_{[E]}(w)$.

3. From Property (4) of Definition 3.2 and the definition of $E \vdash q \diamond w$ it follows immediately that:

$$E \vdash q \diamond w \longrightarrow E.q.w \simeq E.w.q \tag{1}$$

Also, from Definition 4.1(1):

$$p \in I_{[E.q]}(w) \Leftrightarrow E.q.w \simeq E.q.p.w' \text{ (for some } w') \tag{2}$$

$$p \in I_{[E]}(w) \Leftrightarrow E.w \simeq E.p.w'' \text{ (for some } w'') \tag{3}$$

We will assume that p is in one set and show that p is also in the other:

(i) (Forward direction) Assume $p \in I_{[E.q]}(w)$. From (2), the events in $dom_{[E.q]}(p.w')$ are the same as in $dom_{[E.q]}(w)$ and they have the same (non-)dependency with $next_{[E]}(q)$ because $E \vdash q \diamond w$.
Therefore:

$$E \vdash q \diamond (p.w') \longrightarrow E.q.p.w' \simeq E.p.w'.q \qquad \text{similar to (1)} \tag{4}$$

Then:

$$\begin{aligned} p \in I_{[E.q]}(w) &\longrightarrow E.q.w \simeq E.q.p.w' && \text{by (2)} \\ &\longrightarrow E.w.q \simeq E.p.w'.q && \text{by (1) and (4)} \\ &\longrightarrow E.w \simeq E.p.w' && \text{by Property (6) of Def. 3.2} \\ &\longrightarrow p \in I_{[E]}(w) && \text{by (3)} \end{aligned}$$

(ii) (Reverse direction) Assume $p \in I_{[E]}(w)$. From $E \vdash q \diamond w$ it follows that $E.w.q$ is an execution sequence, therefore:

$$\begin{aligned} p \in I_{[E]}(w) &\longrightarrow E.w \simeq E.p.w'' && \text{by (3)} \\ &\longrightarrow E.w.q \simeq E.p.w''.q && \text{by Property (6) of Def. 3.2} \end{aligned} \tag{5}$$

The events in $dom_{[E.q]}(p.w'')$ are the same as $dom_{[E.q]}(w)$ and they have the same (non-)dependency with $next_{[E]}(q)$ because $E \vdash q \diamond w$. Therefore:

$$E \vdash q \diamond (p.w'') \longrightarrow E.q.p.w'' \simeq E.p.w''.q \qquad \text{similar to (1)} \tag{6}$$

Continuing from (5):

$$\begin{aligned} (5) = E.w.q &\simeq E.p.w''.q \\ &\longrightarrow E.q.w \simeq E.q.p.w'' && \text{by (1) and (6)} \\ &\longrightarrow p \in I_{[E.q]}(w) && \text{by (2)} \end{aligned}$$

This concludes the proof of all cases of the lemma. □